JIMWLK and BFKL from your Group I, II, III and 4rho files: summary

This is the cover page for the two full documents:

- `Checks_JIMWLK_BFKL.pdf`: every step of your files, the split into $\log \frac{k^+}{\Lambda}$ and $\log \frac{\vee}{k^+}$ terms row by row, the symmetrization, and the comparison with your JIMWLK and BFKL formulas;
- `Remaining_RC_DGLAP_finite.pdf`: the left-over terms, running coupling, DGLAP and finite terms.

Blue = new or corrected, ~~red~~ = remove.

1 Inputs

- `Group_I.tex`, `Group_III.tex`, `4rho.tex`: identical to the files checked on Oct 31.
- `Group_II.tex`: identical except Eqs. (1.13) ($k\{+\} \to k^+$) and (1.16) (phase of the second term). These are the corrections M2 suggested last time. Eq. (1.16) now passes: 1.9×10^{-16}, previously 0.35.
- Groups I and III are taken with their c.c., Group II without.
- $\vee$ is exact; $\Lambda \to 0$ only where nothing depends on it.

2 How the ρ's are symmetrized

The ρ's do not commute, $[\rho^a(u), \rho^b(v)] = if^{abc}\rho^c(u)\delta(u-v)$. Every term with three or four charges is:

1. summed over all placements of its charges, plus its c.c. for Groups I and III;
2. rewritten as the totally symmetric product plus commutators.

The symmetric three- and four-ρ parts must cancel among themselves. The commutators turn three ρ's into two (and four into three, then two). These two-ρ remainders are added to the genuine two-ρ terms, and only then compared with JIMWLK⊗LO and BFKL⊗LO.

3 Results

Each row is split exactly as

$$\frac{d^3N_{\text{row}}}{d^3k} = \log\frac{k^+}{\Lambda}\,\mathfrak{T}_{\text{row}} + \log\frac{\vee}{k^+}\,\mathfrak{P}_{\text{row}} + R_{\text{row}}.$$

The 79 evaluable terms agree with their three parts to 1.5×10^{-14}.

check	result
four ρ's (`4rho.tex`), symmetric part	cancel, 1.3×10^{-16}
three ρ's $\propto \log \frac{k^+}{\Lambda}$ (22 terms)	cancel, 7.7×10^{-16} (needs F3)
three ρ's $\propto \log \frac{\vee}{k^+}$ (35 terms)	cancel, 1.5×10^{-16}
two ρ's $\propto \log \frac{k^+}{\Lambda}$ vs JIMWLK⊗LO	equal: 0 of 154 colour structures differ
two ρ's $\propto \log \frac{\vee}{k^+}$ vs BFKL⊗LO	equal: 0 of 109 colour structures differ
your BFKL formula (this message)	**correct as written**, 8.2×10^{-15}
your JIMWLK formula (this message)	off by 0.22–0.47 as sent; 2.1×10^{-15} after M2′
poles $\times \log \frac{k^+}{\Lambda}$ (G I Row II : G I Row V 2b UV : G II Row IV)	$1 : -2 : 1$, cancel (8.2×10^{-16})
BFKL terms	pole free

So

$$\frac{d^3N_{\rm LL}}{d^3k} = \mathcal{N}\Big[\log\frac{k^+}{\Lambda}\ \text{JIMWLK}\otimes\text{LO} + \log\frac{\vee}{k^+}\ \text{BFKL}\otimes\text{LO}\Big], \qquad \mathcal{N} = \frac{1}{(2\pi)^3}\frac{g^4}{16\pi^5}\frac{1}{k^+}.$$

Where the BFKL terms sit. Most are in the explicit logs of the rows. In Group III Rows III, V, VI and Group II Row IV they sit inside line 1 (the p^+ or ξ integral), and three exact identities bring them out:

1. the subtraction of your plus prescription cancels the $\log\frac{\vee}{k^+}$ of line 2;
2. the Fourier transform at $y' = z$ gives the (B) terms;
3. the Fourier transform of $\log(y'-y)^2$ gives $\log\frac{\vee}{k^+}H_4$.

4 What still has to change in your files

These are the same as on Oct 31; none of them are in the new uploads yet.

4ρ `4rho.tex` Eq. (1.12) (two ρ, first part): the factor $\big[\log\frac{k^+}{\Lambda} + \log\frac{\vee}{k^+}\big]$ is missing; its value is exactly $1/\log\frac{\vee}{\Lambda}$ of the right one. Two more fixes:

- **F1**: drop ~~$-8N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x'-w)K(y,x',z)\rho\rho UU$~~ in the first part;
- **F2**: second part, $dN^{(1)}$ term 2: the index b' occurs four times; the correct colour factor is $f^{abc}f^{deg}\rho^{b'}(x')\rho^c(y)U^{ag}(y)U^{db'}(w')U^{eb}(z)$.

With all three the lists agree with the checked ones to 5.6×10^{-15}.

F3 Group I Row III, d^3N_{3a}: the log line with the edge kernel must be replaced by $\mathcal{D}$. Without this the three-ρ terms $\propto\log\frac{k^+}{\Lambda}$ leave 0.59.

M3 Group III Row III, first (unnumbered) line: the prefactor is $-\frac{4}{(2\pi)^3}$. Eqs. (1.5)–(1.10) are right.

M4 Group III Row III part 2, measures: $\int_{w,w',x',y',z,x}$, then $\int_{w,x',y',z,x}$ after the w' integral.

M5, M6 Group I Eq. (1.6): $\int_{w,w',x'}$. Unbalanced `\Bigg\{ \Bigg\}` in Group I (1.9), (1.19), (1.20), (1.22), (1.24). Missing commas in $\int_{x,z,y,x',y',w'}$. Group II repeats "We write it in two parts".

M2′ Your JIMWLK formula, as in this message:

- in a term $\mathcal{K}(u,v,t)\,\mathrm{Tr}\{T^eU^\dagger(v)U(u)T^e\,M\}$ the bracket must be $M = (U^\dagger(u)-U^\dagger(t))(U(v)-U(t))$, i.e. at the two points the charges act on, not always at x, y;
- $dN^{(5)}$ has an overall $+$ sign.

The corrected formula is written out in `Checks_JIMWLK_BFKL.pdf`, Sec. "The targets". Your BFKL formula needs no change.

5 Left-over terms

After JIMWLK and BFKL, `Remaining_RC_DGLAP_finite.pdf` lists the remaining terms row by row. They give:

- **Running coupling**:
  - the β_0 of Group I Row II part 1 cancels that of Group I Row V 2b UV exactly;
  - the $\overline{MS}$ coupling counterterm supplies the $\frac{\beta_0}{2}\delta(1-\xi)$ of P_{gg};
  - what remains is the RG term $\frac{\alpha_s\beta_0}{4\pi}\log\frac{\mu_R^2}{\mu_F^2}$LO.

  Equivalently, splitting $\frac{\beta_0}{2}\delta(1-\xi)$ off Group II Row IV (as in your `Other_terms.tex`, label RC) gives LO with $\alpha_s\to\alpha_s(X^2)$; see `RowIV_derivation.pdf`.
- **DGLAP**: $-\frac{\alpha_s}{2\pi}\int_{k^+/\vee}^1\frac{d\xi}{\xi^2}P_{gg}\,\mathrm{LO}_\xi[\ln\frac{X^2\mu_F^2}{c_0^2}]$, with the BFKL region subtracted (Sec. 5 below). The pole goes into $D_{h/g}$.
- **Finite terms**, including $\frac{\alpha_s}{2\pi}N_c(\frac{67}{9}-\frac{\pi^2}{3})$LO and the linear-polarization term.

Open points: Q1 (pole $\propto\log(1-\frac{k^+}{\vee})$, vanishes for $\vee\to\infty$), Q2 (finite four-ρ term not in your files), Q3 (finite part of the $\mathcal{D}$ region).

6 Your `Other_terms.tex`

27 of its 30 equations are identical, token by token, to the version verified on Nov 3. The other three:

1. **Group I Row II part 1**: $\log(1-\frac{\Lambda}{k^+})$ dropped. Correct for $\Lambda \to 0$ (it is $O(\Lambda/k^+)$).

2. **Group I Row V, 2b finite**: limits $\int_\Lambda^{k^+-\Lambda} \to \int_0^{k^+}$ and $\int_{k^+-\Lambda}^{\Lambda} \to \int_{k^+}^0$. Correct for $\Lambda \to 0$ (the plus-prescribed integrands are integrable).

3. **Group II Row IV**: only the labels DGLAP / RC are new. One thing in it must still be fixed, the term your advisor objected to:

$$-2N_c \frac{1-\xi}{\xi} \log\left(\frac{(y'-y)^2\mu^2}{4}\right) \frac{d^3N_{\xi\,\mathrm{LO}}}{d^3k}^{y} .$$

   - In $\frac{d^3N_{\xi\,\mathrm{LO}}}{d^3k}^{y}$ the only y' dependence is the phase $e^{-ik\cdot(y'-y)/\xi}$. With the log written *outside*, $\int_{y'} e^{-ik\cdot(y'-y)/\xi} = (2\pi\xi)^2\delta^{(2)}(\mathbf{k}) = 0$, so the term vanishes.
   - With the log *inside* the y' integral, $\int_{y'} e^{-ik\cdot(y'-y)/\xi} \log\frac{(y'-y)^2\mu^2}{4} = -\frac{4\pi\xi^2}{\mathbf{k}^2}$, and it is exactly right.
   - Write it either way:

$$-2N_c \frac{1-\xi}{\xi} \frac{d^3N_{\xi\,\mathrm{LO}}}{d^3k}^{y} \left[\log\frac{(y'-y)^2\mu^2}{4}\right] = +\frac{8\pi N_c\,\xi(1-\xi)}{\mathbf{k}^2} \frac{d^3N_{\mathrm{LO}}}{d^3k}^{y} ,$$

     with $[\dots]$ meaning "inside the integrals", and $\frac{d^3N_{\mathrm{LO}}}{d^3k}^{y}$ having no y' at all.
   - The second form is symmetric (both gluon points at one z), so the $y \leftrightarrow y'$ asymmetry your advisor raised is gone.
   - Its ξ integral is $[\log\frac{\vee}{k^+} - (1-\frac{k^+}{\vee})]H_4$. This is what cancels the BFKL $\log\frac{\vee}{k^+}H_4$ taken out of DGLAP and leaves your $-(1-\frac{k^+}{\vee})H_4$ line.

   Everything else in Group II Row IV (the $-(1-\frac{k^+}{\vee})H_4$ term and the linear-polarization line with $d_\perp = 2$) is correct.